

复合函数之偏导数

$z=f(y)$, $y=g(x)$, 可导, 则 $f(g(x))$ 也可导.

$$\frac{dz}{dx} = f'(g(x)) \cdot g'(x) \quad \downarrow \text{链式法则}$$

~~二元函数~~ $z=f(x,y)$, $x=g(u,v)$, $y=h(u,v)$ 可微

则 $z=f(g(u,v), h(u,v))$ 可微

$z=f(x,y)$ 看成 (x,y) 之函数

$$dz = f_x dx + f_y dy$$

$z=f(g(u,v), h(u,v))$ 看成复合函数, 是 (u,v) 之函数

$$dz = \frac{\partial f}{\partial u} du + \frac{\partial f}{\partial v} dv$$

$x=g(u,v)$, $y=h(u,v)$: 是 (u,v) 之可微函数

$$dx = g_u du + g_v dv$$

$$dy = h_u du + h_v dv$$

$$\Rightarrow dz \stackrel{?}{=} f_x (g_u du + g_v dv) + f_y (h_u du + h_v dv)$$

仍为复合函数 $z=f(g(u,v), h(u,v))$ 是 (u,v) 之函数

$$dz = \frac{\partial f}{\partial u} du + \frac{\partial f}{\partial v} dv$$

$$= (\underbrace{f_x \cdot g_u}_{\text{red circle}} + \underbrace{f_y \cdot h_u}_{\text{blue circle}}) du + (\underbrace{f_x \cdot g_v}_{\text{red circle}} + \underbrace{f_y \cdot h_v}_{\text{blue circle}}) dv$$

$$= f_x(\underline{g_u du + g_v dv}) + f_y(h_u du + h_v dv)$$

$$= f_x dx + f_y dy$$

$\underline{z}$ 是 $\underline{x}, \underline{y}$ 的充分微分不变量,

若 $z = f(x, y)$ 看成 (x, y) 的函数

$$dz = f_x dx + f_y dy$$

独立变量: dx, dy : 与 x, y 无关的独立变量

$$d^2 z = d(dz) = (dz)_x dx + (dz)_y dy$$

$$\begin{cases} (dz)_x = (f_x dx + f_y dy)_x = f_{xx} dx + f_{yx} dy \\ (dz)_y = f_{xy} dx + f_{yy} dy \end{cases}$$

$$\begin{aligned} d^2 z &= (f_{xx} dx + f_{yx} dy) dx + (f_{xy} dx + f_{yy} dy) dy \\ &= f_{xx} dx^2 + 2 f_{xy} dx dy + f_{yy} dy^2 \end{aligned}$$

作复合函数 $z = f(g(u, v), h(u, v))$ 令 (u, v) 是独立变量

$$dz = \frac{\partial f}{\partial u} du + \frac{\partial f}{\partial v} dv$$

$$= f_x dx + f_y dy$$

$$\begin{cases} dx = g_u(u, v) du + g_v(u, v) dv \\ dy = h_u(u, v) du + h_v(u, v) dv \end{cases}$$

$$d^2 z = (dz)_u du + (dz)_v dv$$

$$dz = f_x(g(u,v), h(u,v)) \underbrace{dx}_{\frac{\partial}{\partial}(u,v) \text{ 的 主 部 }} + f_y(g(u,v), h(u,v)) dy$$

$$(dz)_u = (f_{xx} \cdot g_u + f_{xy} \cdot h_u) dx + f_x(g, h) [g_{uu} du + g_{vu} dv]$$

$$+ (f_{yx} g_u + f_{yy} \cdot h_u) dy$$

$$+ f_y(g, h) \cdot [h_{uu} du + h_{vu} dv]$$

$$\neq f_{xx} dx^2 + 2 f_{xy} dx dy + f_{yy} dy^2 \quad \leftarrow \text{这里多了两项!}$$

$$dx = g_u du + g_v dv \quad dy = h_u du + h_v dv$$

二阶结论: 不再用备微分式之不变性.

例: 设 $f(x, y, z)$ 是二阶连续可微函数 $y = \varphi(x) \in C^2(\mathbb{R})$

$u = u(x, z) = f(x, \varphi(x), z)$ 是二阶连续可微, $\frac{\partial u}{\partial x}, \frac{\partial^2 u}{\partial x^2}$.

$$\frac{\partial u}{\partial x} = f_x(x, \varphi(x), z) + f_y(x, \varphi(x), z) \cdot \varphi'(x)$$

$$\frac{\partial^2 u}{\partial x^2} = f_{xx}(x, \varphi(x), z) + f_{xy} \cdot \varphi'(x)$$

$$+ f_{yx}(x, \varphi(x), z) \varphi'(x) + f_{yy} \cdot \varphi'(x) \cdot \varphi'(x)$$

$$+ f_y(x, \varphi(x), z) \cdot \varphi''(x)$$

$$f_x(x, \varphi(x), z):$$

↑ 仅对第一变量

求偏导数

$$= f_{xx} + 2f_{xy} \varphi'(x) + f_{yy} (\varphi'(x))^2 + f_y \cdot \varphi''(x),$$

例 设 $u(x, y, z)$ 是调和函数. 即: $\Delta u = 0$ (调和函数)

$$u_{xx} + u_{yy} + u_{zz} = 0 \quad (\text{调和方程})$$

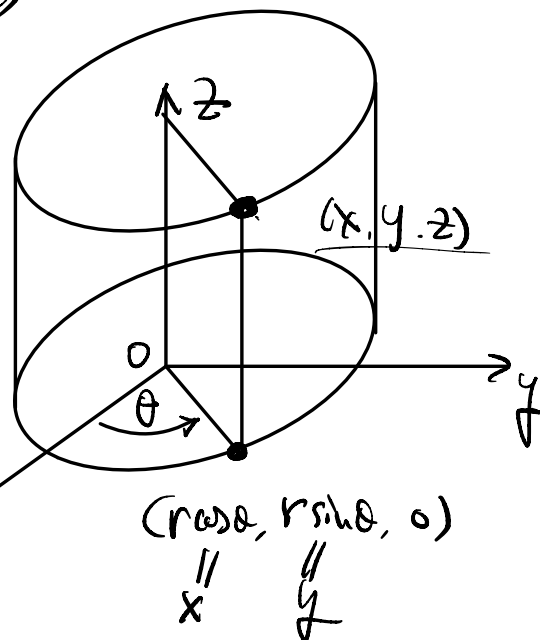
则在柱坐标变换

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \\ z = z \end{cases}$$

则 $u = u(r \cos \theta, r \sin \theta, z)$ 满足:

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} + \frac{\partial^2 u}{\partial z^2} = 0.$$

(柱坐标下调和方程)



$$u = u(r \cos \theta, r \sin \theta, z)$$

证明: $u_r = u_x \cos \theta + u_y \sin \theta$

$$u_{rr} = u_{xx} \cos \theta \cos \theta + u_{xy} \sin \theta \cos \theta$$

$$+ u_{yx} \cos \theta \sin \theta + u_{yy} \sin \theta \sin \theta$$

$$u_\theta = u_x \cdot r(-\sin \theta) + u_y \cdot r \cos \theta$$

$$u_{\theta\theta} = u_{xx} (r(-\sin \theta))^2 + u_{xy} r \cos \theta r(-\sin \theta) - r u_x \cdot \cos \theta$$

$$+ u_{yx} r(-\sin \theta) \cdot r \cos \theta + u_{yy} (r \cos \theta)^2 + r u_y \cdot (-\sin \theta)$$

$$\frac{1}{r^2} u_{\theta\theta} = u_{xx} \cdot \cancel{r^2 \sin^2 \theta} - 2 u_{xy} \cancel{r^2 \sin \theta \cos \theta} + \underline{u_{yy} \cdot r^2 \cos^2 \theta}$$

$$- \cancel{r} (u_x \cos \theta + u_y \sin \theta)$$

$$u_r = u_x \cos \theta + u_y \sin \theta$$

$$+) u_{rr} = u_{xx} \cos^2 \theta + 2 u_{xy} \sin \theta \cos \theta + u_{yy} \sin^2 \theta$$

$$u_{rr} + \frac{1}{r^2} u_{\theta\theta} = u_{xx} + u_{yy} - \frac{1}{r} u_r$$

$$\therefore u_{xx} + u_{yy} = u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta} \quad \leftarrow$$

$$u(x, y, z) \stackrel{0}{=} \nabla^2 u = 0:$$

$$0 = \underbrace{u_{xx} + u_{yy}} + \underbrace{u_{zz}} = u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta} + u_{zz}$$

$\therefore$ 在柱坐标下, u 满足:

$$u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta} + u_{zz} = 0.$$

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柱坐标

直角坐标

$$(r, \theta, z) \longleftrightarrow (x, y, z):$$

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \\ z = z \end{cases}$$

§3. 隐函数与偏导数

一. 单个方程情形:

定义3.1 ① 一元隐函数: 由 $F(x, y) = 0$ 决定之函数

$$y = y(x), x \in I \subset \mathbb{R} \Leftrightarrow F(x, y(x)) \equiv 0, x \in I.$$

$y = y(x)$ 称为由 $F(x, y) = 0$ 决定之一元隐函数.

② 二元隐函数: 由 $F(x, y, z) = 0$ 决定之函数

$$z = z(x, y): (x, y) \in E \subset \mathbb{R}^2 \Leftrightarrow F(x, y, z(x, y)) \equiv 0, (x, y) \in E.$$

$z = z(x, y)$ 称为由 $F(x, y, z) = 0$ 决定之二元隐函数.

隐函数与导数:

设 $F(x, y)$ 可微 隐函数 $y = y(x)$ 也可微:

$\Rightarrow F(x, y(x))$ 可微. $F(x, y(x)) \equiv 0$. 两边对 x 求导:

$$F_x + F_y \cdot y'(x) = 0$$

$$y'(x) = -\frac{F_x(x, y)}{F_y(x, y)} \quad (\text{当 } F_y(x, y) \neq 0).$$

二元隐函数与偏导数:

设 $F(x, y, z)$ 可微 隐函数 $z = z(x, y)$ 也可微:

$\Rightarrow F(x, y, z(x, y))$ 可微:

$F(x, y, z(x, y)) \equiv 0$: 两边关于 x 求偏导数

$$z_x = - \frac{F_x(x, y, z)}{F_z(x, y, z)} \quad \left(\frac{\partial}{\partial z} F_z(x, y, z) \neq 0 \right)$$

同理:

$$z_y = - \frac{F_y(x, y, z)}{F_z(x, y, z)}$$

二. 方程组 - 隐函数:

$$\text{由方程组} \begin{cases} F_1(x, y, z) = 0 \\ F_2(x, y, z) = 0 \end{cases} \text{ 满足} \rightarrow \text{函数组} \begin{cases} y = y(x), \\ z = z(x), \end{cases} x \in I$$

称为隐函数(组):

$$\begin{cases} F_1(x, y(x), z(x)) \equiv 0, \\ F_2(x, y(x), z(x)) \equiv 0, \end{cases} \quad x \in I$$

如果 $F_k(x, y, z)$ ($k=1, 2$) 可微, 且 $\begin{cases} y = y(x) \\ z = z(x) \end{cases}$ 也可微

$$\Rightarrow \begin{cases} F_1(x, y(x), z(x)) \\ F_2(x, y(x), z(x)) \end{cases} \text{ 可微}$$

$$\begin{cases} F_1(x, y(x), z(x)) \equiv 0 \\ F_2(x, y(x), z(x)) \equiv 0 \end{cases} \quad \text{两边对 } x \text{ 求导}$$

$$\Rightarrow \begin{cases} F_{1x} + F_{1y} \cdot y'(x) + F_{1z} \cdot z'(x) \equiv 0 \\ F_{2x} + F_{2y} \cdot y'(x) + F_{2z} \cdot z'(x) \equiv 0 \end{cases}$$

是关于 $y'(x), z'(x)$ 的方程组.

$\exists$ 解 $y'(x), z'(x) \Leftrightarrow$ 系数行列式 $\neq 0$

$$\text{系数行列式 } J \triangleq \begin{vmatrix} F_{1y} & F_{1z} \\ F_{2y} & F_{2z} \end{vmatrix} \triangleq \frac{\partial(F_1, F_2)}{\partial(y, z)}$$

称为 Jacobi 行列式

由 Cramer 法则: $\frac{1}{2} J \neq 0$ 时 $\Rightarrow \exists$ 解 $\begin{cases} y'(x) \\ z'(x) \end{cases}$

$$\begin{cases} F_{1y} \cdot y'(x) + F_{1z} z'(x) = -F_{1x} \\ F_{2y} \cdot y'(x) + F_{2z} z'(x) = -F_{2x} \end{cases}$$

$$\begin{cases} F_{2y} \cdot y'(x) + F_{2z} z'(x) = -F_{2x} \end{cases}$$

$$\therefore y'(x) = -\frac{1}{J} \begin{vmatrix} F_{1x} & F_{1z} \\ F_{2x} & F_{2z} \end{vmatrix} = -\frac{1}{J} \frac{\partial(F_1, F_2)}{\partial(x, z)} = -\frac{\frac{\partial(F_1, F_2)}{\partial(x, z)}}{\frac{\partial(F_1, F_2)}{\partial(y, z)}}$$

$$z'(x) = -\frac{1}{J} \begin{vmatrix} F_{1y} & F_{1x} \\ F_{2y} & F_{2x} \end{vmatrix} = \frac{1}{J} \begin{vmatrix} F_{1x} & F_{1y} \\ F_{2x} & F_{2y} \end{vmatrix} = \frac{\frac{\partial(F_1, F_2)}{\partial(x, y)}}{\frac{\partial(F_1, F_2)}{\partial(y, z)}}$$

$$\text{改写: } y'(x) = \frac{1}{J} \begin{vmatrix} F_{1z} & F_{1x} \\ F_{2z} & F_{2x} \end{vmatrix} = \frac{\frac{\partial(F_1, F_2)}{\partial(z, x)}}{\frac{\partial(F_1, F_2)}{\partial(y, z)}}$$

换位置:
相当于交换列,
多出负号。

$$J = \frac{\partial(F_1, F_2)}{\partial(y, z)} \neq 0 \Rightarrow \exists y'(x), z'(x).$$

$$y'(x): \frac{\partial(F_1, F_2)}{\partial(z, x)}$$

$$z'(x): \frac{\partial(F_1, F_2)}{\partial(x, y)}$$

$x, y, z, x, y, z, x, y, z$
对称轮换:

作业. P124. 2, 4, 5, 6, 9.